

Eigenvalues and inverse problems

Find a near-largest nonnormal eigenvalue using few matrix-vector products

Difficulty: challenging **Importance:** interesting to the community**Status:** Solution claimed**Topic:** nonsymmetric eigenvalue computation; query complexity

Rating rationale: Challenging reflects a query bound for general nonnormal matrices, whose eigenvalues need not be controlled by power iteration; community impact is the complexity of large nonsymmetric eigenproblems.

Resolution claim — 2026-09-11

Affirmative resolution claimed; submitted by Matthew Colbrook. See the [complete manuscript](#), **Theorem KE-03, sections 1–5** ([PDF](#) · [LaTeX](#)), prepared 11 September 2026.

The manuscript claims an algorithm using $O(\varepsilon^{-2}[1+\log(nK)])$ exact matrix-vector queries, with success probability at least 0.997, for every input in the displayed model. It supplies both eigenvalue-location guarantees using the given condition bound K and finite exact arithmetic between queries. This is a query-count claim; it does not bound total runtime, bit complexity or floating-point error.

The manuscript discloses generation in a ChatGPT conversation. Its full proof has not been independently verified here, and no publication or priority claim is made. [Supporting diagnostics](#) were rerun successfully; these are not an independent proof audit. This update checks the submitted claim against the displayed target; the dated literature checks below remain the record of the earlier search. The ratings above are historical, and this entry is excluded from the open count.

Context and notation

All norms are Euclidean vector norms or their induced matrix norms.

Problem statement

An unknown diagonalizable $A \in \mathbb{C}^{n \times n}$ is available only through exact queries $v \mapsto Av$. A number $K \geq 1$ is supplied, with the promise that $A = V\Lambda V^{-1}$ for some diagonal Λ and $\|V\|_2\|V^{-1}\|_2 \leq K$. Assume $\rho(A) = \max_{\mu \in \sigma(A)} |\mu| > 0$; neither the eigenvalues nor V are supplied.

Do universal constants $C, a, b > 0$ and a randomized algorithm exist that, for every such input and every $\varepsilon \in (0, 1/2)$, returns a complex number z with probability at least 0.99 such that

$$\text{there exists } \mu \in \sigma(A) : \quad |\mu| \geq (1 - \varepsilon)\rho(A), \quad |z - \mu| \leq \varepsilon\rho(A),$$

using at most $C[1 + \log(nK)]^a \varepsilon^{-b}$ matrix-vector queries? The algorithm may perform finite exact arithmetic between queries; the requested bound concerns query count. Products with A^* and solutions of shifted linear systems are not additional oracle operations. Supplying K and fixing a success probability make the computational formulation explicit.

References

Amsel et al., [workshop report](#), Problem 3.9. Shah, Srivastava, and Zeng, *Sparse Pseudospectral Shattering*, [first version](#), §1.3, Theorem 1.5 and its power-iteration argument, supplies the related spectral-radius estimate cited by the workshop.

Earlier status check — 2026-09-08

Searches for *nonnormal eigenvalue query complexity 2026* and *nonnormal spectral radius matrix-vector algorithm 2026* found no resolution. The radius estimate controls a nonnegative scalar and allows backward error; it does not locate a near-extremal eigenvalue in the complex plane as required here. The [April 2026 third version](#) replaces that application with a GMRES application (§6), so the version-specific locator above is intentional. The workshop still poses Problem 3.9 in its August update, separately from its resolved general Ritz-compression question.

Audit update — 2026-09-10

Rechecked [workshop version 3](#), Problem 3.9. Searches for nonnormal largest-eigenvalue matrix-vector query bounds found no theorem or lower bound settling the displayed question. Hermitian or singular-value oracle results do not answer it.